



Keywords

Views

Collections

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This guide shows the basic Keyword, the Views and Collections of Basic4Android.

To search for a given word or sentence use the Search function in the Edit menu.

To learn the Basic4Android basics, look at the [Beginner's Guide](#).

A more advanced guide can be downloaded [User's Guide](#).

1 B4A Keywords

List of types:

[Bit](#)
[DateTime](#)
[Exception](#)
[Keywords](#)
[LayoutValues](#)
[String](#)
[StringBuilder](#)
[Timer](#)

1.1 Bit

Bit is a predefined object containing bitwise related methods.

Example:

```
Dim flags As Int
flags = Bit.Or(100, 200)
```

Events:

None

Members:

- 🔧 [And](#) (N1 As Int, N2 As Int) As Int
- 🔧 [Not](#) (N As Int) As Int
- 🔧 [Or](#) (N1 As Int, N2 As Int) As Int
- 🔧 [ParseInt](#) (Value As String, Radix As Int) As Int
- 🔧 [ShiftLeft](#) (N As Int, Shift As Int) As Int
- 🔧 [ShiftRight](#) (N As Int, Shift As Int) As Int
- 🔧 [ToBinaryString](#) (N As Int) As String
- 🔧 [ToHexString](#) (N As Int) As String
- 🔧 [ToOctalString](#) (N As Int) As String
- 🔧 [UnsignedShiftRight](#) (N As Int, Shift As Int) As Int
- 🔧 [Xor](#) (N1 As Int, N2 As Int) As Int

Members description:

- 🔧 [And](#) (N1 As Int, N2 As Int) As Int

Returns the bitwise AND of the two values.

📌 **Not (N As Int) As Int**

Returns the bitwise complement of the given value.

📌 **Or (N1 As Int, N2 As Int) As Int**

Returns the bitwise OR of the two values.

📌 **ParseInt (Value As String, Radix As Int) As Int**

Parses Value as an integer using the specified radix.

Radix - Should be between 2 to 36.

📌 **ShiftLeft (N As Int, Shift As Int) As Int**

Shifts N left.

Shift - Number of positions to shift.

📌 **ShiftRight (N As Int, Shift As Int) As Int**

Shifts N right.

Keeps the original value sign

Shift - Number of positions to shift.

📌 **ToBinaryString (N As Int) As String**

Returns a string representation of N in base 2.

📌 **ToHexString (N As Int) As String**

Returns a string representation of N in base 16.

📌 **ToOctalString (N As Int) As String**

Returns a string representation of N in base 8.

📌 **UnsignedShiftRight (N As Int, Shift As Int) As Int**

Shifts N right.

Shifts zeroes in the leftmost positions.

Shift - Number of positions to shift.

📌 **Xor (N1 As Int, N2 As Int) As Int**

Returns the bitwise XOR of the two values.

1.2 DateTime

Date and time related methods.

DateTime is a predefined object. You should not declare it yourself.

Date and time values are stored as ticks. Ticks are the number of milliseconds since January 1, 1970.

This value is too large to be stored in an Int variable. It should only be stored in a Long variable.

The methods `DateTime.Date` and `DateTime.Time` convert the ticks value to a string. You can get the current time with `DateTime.Now`.

Example:

```
Dim now As Long
now = DateTime.Now
Msgbox("The date is: " & DateTime.Date(now) & CRLF & _
    "The time is: " & DateTime.Time(now), "")
```

Events:

None

Members:

 [Add](#) (Ticks As Long, Years As Int, Months As Int, Days As Int) As Long

 [Date](#) (Ticks As Long) As String

 [DateFormat](#) As String

 [DateParse](#) (Date As String) As Long

 [GetDayOfMonth](#) (Ticks As Long) As Int

 [GetDayOfWeek](#) (Ticks As Long) As Int

 [GetDayOfYear](#) (Ticks As Long) As Int

 [GetHour](#) (Ticks As Long) As Int

 [GetMinute](#) (Ticks As Long) As Int

 [GetMonth](#) (Ticks As Long) As Int

 [GetSecond](#) (Ticks As Long) As Int

 [GetYear](#) (Ticks As Long) As Int

 [Now](#) As Long [read only]

 [TicksPerDay](#) As Long

 [TicksPerHour](#) As Long

 [TicksPerMinute](#) As Long

 [TicksPerSecond](#) As Long

 [Time](#) (Ticks As Long) As String

 [TimeFormat](#) As String

TimeParse (Time As String) As Long

Members description:

Add (Ticks As Long, Years As Int, Months As Int, Days As Int) As Long

Returns a ticks value which is the result of adding the specified time spans to the given ticks value.
Pass negative values if you want to subtract the values.

Example:

```
Dim Tomorrow As Long
```

```
Tomorrow = DateTime.Add(DateTime.Now, 0, 0, 1)
```

```
Log("Tomorrow date is: " & DateTime.Date(Tomorrow))
```

Date (Ticks As Long) As String

Returns a string representation of the date (which is stored as ticks).
The date format can be set with the DateFormat keyword.

Example:

```
Log("Today is: " & DateTime.Date(DateTime.Now))
```

DateFormat As String

Gets or sets the format used to parse date strings.

See this page for the supported patterns: [formats](#).

The default pattern is MM/dd/yyyy (04/23/2002 for example).

DateParse (Date As String) As Long

Parses the given date string and returns its ticks representation.

An exception will be thrown if parsing fails.

Example:

```
Dim SomeTime As Long
```

```
SomeTime = DateTime.DateParse("02/23/2007")
```

GetDayOfMonth (Ticks As Long) As Int

Returns the day of month component from the ticks value.

Values are between 1 to 31.

GetDayOfWeek (Ticks As Long) As Int

Returns the day of week component from the ticks value.

Values are between 1 to 7.

GetDayOfYear (Ticks As Long) As Int

Returns the day of year component from the ticks value.

Values are between 1 to 366.

GetHour (Ticks As Long) As Int

Returns the hour of day component from the ticks value.

Values are between 0 to 23.

GetMinute (Ticks As Long) As Int

Returns the minutes within a hour component from the ticks value.
Values are between 0 to 59.

GetMonth (Ticks As Long) As Int

Returns the month of year component from the ticks value.
Values are between 1 to 12.

GetSecond (Ticks As Long) As Int

Returns the seconds within a minute component from the ticks value.
Values are between 0 to 59.

GetYear (Ticks As Long) As Int

Returns the year component from the ticks value.

Now As Long [read only]

Gets the current time as ticks (number of milliseconds since January 1, 1970).

TicksPerDay As Long

TicksPerHour As Long

TicksPerMinute As Long

TicksPerSecond As Long

Time (Ticks As Long) As String

Returns a string representation of the time (which is stored as ticks).
The time format can be set with the TimeFormat keyword.

Example:

```
Log("The time now is: " & DateTime.Date(DateTime.Now))
```

TimeFormat As String

Gets or sets the format used to parse time strings.
See this page for the supported patterns: [formats](#).
The default pattern is HH:mm:ss (23:45:12 for example).

TimeParse (Time As String) As Long

Parses the given date string and returns its ticks representation.
Note that the returned value date will be today.

1.3 Exception

Holds a thrown exception.

You can access the last thrown exception by calling LastException.

For example:

```
Try
```

```
Dim in As InputStream
```

```
in = File.OpenInput(File.DirInternal, "SomeMissingFile.txt")
```

```
' ...
```

```
Catch
```

```
Log(LastException.Message)
End Try
If in.IsInitialized Then in.Close
```

Events:

None

Members:

 IsInitialized As Boolean

 Message As String [read only]

Members description:

 IsInitialized As Boolean

 Message As String [read only]

1.4 Keywords

These are the internal keywords.

Events:

None

Members:

 Abs (Number As Double) As Double

 ACos (Value As Double) As Double

 ACosD (Value As Double) As Double

 Array

 Asc (Char As Char) As Int

 ASin (Value As Double) As Double

 ASinD (Value As Double) As Double

 ATan (Value As Double) As Double

 ATanD (Value As Double) As Double

 ATan2 (Y As Double, X As Double) As Double

 ATan2D (Y As Double, X As Double) As Double

- ⇒ [BytesToString](#) (Data() As Byte, StartOffset As Int, Length As Int, CharSet As String) As String
- ⇒ [CallSub](#) (Component As Object, Sub As String) As String
- ⇒ [CallSub2](#) (Component As Object, Sub As String, Argument As Object) As String
- ⇒ [CallSub3](#) (Component As Object, Sub As String, Argument1 As Object, Argument2 As Object) As String
- ⇒ [CallSubDelayed](#) (Component As Object, Sub As String)
- ⇒ [CallSubDelayed2](#) (Component As Object, Sub As String, Argument As Object)
- ⇒ [CallSubDelayed3](#) (StartIfNecessary As Boolean, Component As Object, Sub As String, Argument1 As Object, Argument2 As Object)
- ⇒ [CancelScheduledService](#) (Service As Object)
- ⇒ [Catch](#)
- ⇒ [cE](#) As Double
- ⇒ [Ceil](#) (Number As Double) As Double
- ⇒ [CharsToString](#) (Chars() As Char, StartOffset As Int, Length As Int) As String
- ⇒ [Chr](#) (UnicodeValue As Int) As Char
- ⇒ [ConfigureHomeWidget](#) (LayoutFile As String, EventName As String, UpdateIntervalMinutes As Int, WidgetName As String, CenterWidget As Boolean) As android.widget.RemoteViews
- ⇒ [Continue](#)
- ⇒ [Cos](#) (Radians As Double) As Double
- ⇒ [CosD](#) (Degrees As Double) As Double
- ⇒ [cPI](#) As Double
- ⇒ [CRLF](#) As String
- ⇒ [Density](#) As Float
- ⇒ [Dim](#)
- ⇒ [DipToCurrent](#) (Length As Int) As Int
- ⇒ [DoEvents](#)
- ⇒ [Exit](#)

⇒ **ExitApplication**

⇒ **False** As Boolean

⇒ **File** As File

⇒ **Floor** (Number As Double) As Double

⇒ **For**

⇒ **GetDeviceLayoutValues** As LayoutValues

⇒ **GetType** (object As Object) As String

⇒ **If**

⇒ **InputList** (Items As List, Title As String, CheckedItem As Int) As Int

⇒ **InputMultiList** (Items As List, Title As String) As List

⇒ **Is**

⇒ **IsBackgroundTaskRunning** (ContainerObject As Object, TaskId As Int) As Boolean

⇒ **IsNumber** (Text As String) As Boolean

⇒ **IsPaused** (Component As Object) As Boolean

⇒ **LastException** As Exception

⇒ **LoadBitmap** (Dir As String, FileName As String) As Bitmap

⇒ **LoadBitmapSample** (Dir As String, FileName As String, MaxWidth As Int, MaxHeight As Int) As Bitmap

⇒ **Log** (Message As String)

⇒ **Logarithm** (Number As Double, Base As Double) As Double

⇒ **Max** (Number1 As Double, Number2 As Double) As Double

⇒ **Me** As Object

⇒ **Min** (Number1 As Double, Number2 As Double) As Double

⇒ **Msgbox** (Message As String, Title As String)

⇒ **Msgbox2** (Message As String, Title As String, Positive As String, Cancel As String, Negative As String, Icon As android.graphics.Bitmap) As Int

⇒ **Not** (Value As Boolean) As Boolean

 **Null** As Object

 **NumberFormat** (Number As Double, MinimumIntegers As Int, MaximumFractions As Int) As String

 **NumberFormat2** (Number As Double, MinimumIntegers As Int, MaximumFractions As Int, MinimumFractions As Int, GroupingUsed As Boolean) As String

 **PerXToCurrent** (Percentage As Float) As Int

 **PerYToCurrent** (Percentage As Float) As Int

 **Power** (Base As Double, Exponent As Double) As Double

 **ProgressDialogHide**

 **ProgressDialogShow** (Text As String)

 **ProgressDialogShow2** (Text As String, Cancelable As Boolean)

 **QUOTE** As String

 **Regex** As Regex

 **Return**

 **Rnd** (Min As Int, Max As Int) As Int

 **RndSeed** (Seed As Long)

 **Round** (Number As Double) As Long

 **Round2** (Number As Double, DecimalPlaces As Int) As Double

 **Select**

 **Sender** As Object

 **Sin** (Radians As Double) As Double

 **SinD** (Degrees As Double) As Double

 **Sqrt** (Value As Double) As Double

 **StartActivity** (Activity As Object)

 **StartService** (Service As Object)

 **StartServiceAt** (Service As Object, Time As Long, DuringSleep As Boolean)

 **StopService** (Service As Object)

🔧 Sub

🔧 SubExists (Object As Object, Sub As String) As Boolean

🔧 TAB As String

🔧 Tan (Radians As Double) As Double

🔧 TanD (Degrees As Double) As Double

🔧 ToastMessageShow (Message As String, LongDuration As Boolean)

🔧 True As Boolean

🔧 Try

🔧 Type

🔧 Until

🔧 While

Members description:

🔧 Abs (Number As Double) As Double

Returns the absolute value.

🔧 ACos (Value As Double) As Double

Returns the angle measured in radians.

🔧 ACosD (Value As Double) As Double

Returns the angle measured in degrees.

🔧 Array

Creates a single dimension array of the specified type.

The syntax is: Array As type (list of values).

Example:

```
Dim Days() As String
Days = Array As String("Sunday", "Monday", ...)
```

🔧 Asc (Char As Char) As Int

Returns the unicode code point of the given character or first character in string.

🔧 ASin (Value As Double) As Double

Returns the angle measured in radians.

🔧 ASinD (Value As Double) As Double

Returns the angle measured in degrees.

ATan (Value As Double) As Double

Returns the angle measured in radians.

ATanD (Value As Double) As Double

Returns the angle measured in degrees.

ATan2 (Y As Double, X As Double) As Double

Returns the angle measured with radians.

ATan2D (Y As Double, X As Double) As Double

Returns the angle measured with degrees.

BytesToString (Data() As Byte, StartOffset As Int, Length As Int, CharSet As String) As String

Decodes the given bytes arrays as a string.

Data - The bytes array.

StartOffset - The first byte to read.

Length - Number of bytes to read.

CharSet - The name of the character set.

Example:

```
Dim s As String
s = BytesToString(Buffer, 0, Buffer.Length, "UTF-8")
```

CallSub (Component As Object, Sub As String) As String

Calls the given sub. CallSub can be used to call a sub which belongs to a different module.

However the sub will only be called if the other module is not paused. In that case an empty string will be returned.

You can use IsPaused to test whether a module is paused.

This means that one activity cannot call a sub of a different activity. As the other activity will be paused for sure.

CallSub allows an activity to call a service sub or a service to call an activity sub.

Note that it is not possible to call subs of code modules.

CallSub can also be used to call subs in the current module. Pass an empty string as the component in that case.

Example:

```
CallSub(Main, "RefreshData")
```

CallSub2 (Component As Object, Sub As String, Argument As Object) As String

Similar to CallSub. Calls a sub with a single argument.

**CallSub3 (Component As Object, Sub As String, Argument1 As Object, Argument2 As Object) As String**

Similar to CallSub. Calls a sub with two arguments.

CallSubDelayed (Component As Object, Sub As String)

CallSubDelayed is a combination of StartActivity, StartService and CallSub.

Unlike CallSub which only works with currently running components, CallSubDelayed will first start the target component if needed.

CallSubDelayed can also be used to call subs in the current module. Instead of calling these subs directly, a message will be sent to the message queue.

The sub will be called when the message is processed. This is useful in cases where you want to do something "right after" the current sub (usually related to UI events).

Note that if you call an Activity while the whole application is in the background (no visible activities), the sub will be executed once the target activity is resumed.

CallSubDelayed2 (Component As Object, Sub As String, Argument As Object)

Similar to CallSubDelayed. Calls a sub with a single argument.



CallSubDelayed3 (StartIfNecessary As Boolean, Component As Object, Sub As String, Argument1 As Object, Argument2 As Object)

Similar to CallSubDelayed. Calls a sub with two arguments.

CancelScheduledService (Service As Object)

Cancels previously scheduled tasks for this service.

Catch

Any exception thrown inside a try block will be caught in the catch block.

Call LastException to get the caught exception.

Syntax:

Try

...

Catch

...

End Try

cE As Double

e (natural logarithm base) constant.

Ceil (Number As Double) As Double

Returns the smallest double that is greater or equal to the specified number and is equal to an integer.

CharsToString (Chars() As Char, StartOffset As Int, Length As Int) As String

Creates a new String by copying the characters from the array.

Copying starts from StartOffset and the number of characters copied equals to Length.

Chr (UnicodeValue As Int) As Char

Returns the character that is represented by the given unicode value.



ConfigureHomeWidget (LayoutFile As String, EventName As String, UpdateIntervalMinutes As Int, WidgetName As String, CenterWidget As Boolean) As android.widget.RemoteViews

Creates a RemoteViews object based on the layout file. The compiler will generate the required XML files based on the parameters.

See the widgets tutorial for more information: [Widgets tutorial](#).

Note that all parameters must be strings or numbers as they are read by the compiler.

LayoutFile - The widget layout file.

EventName - Sets the subs that will handle the RemoteViews events.

UpdateIntervalMinutes - Sets the update interval. Pass 0 to disable automatic updates. Minimum value is 30 (or 0).

WidgetName - The name of the widget as appears in the widgets list.

CenterWidget - Whether to center the widget.

Continue

Stops executing the current iteration and continues with the next one.

Cos (Radians As Double) As Double

Calculated the trigonometric cosine function. Angle measured in radians.

CosD (Degrees As Double) As Double

Calculated the trigonometric cosine function. Angle measured in degrees.

cPI As Double

PI constant.

CRLF As String

New line character. The value of Chr(13).

Density As Float

Returns the device scale, which is DPI / 160.
(DPI stands for dots per inch).

Dim

Declares a variable.

Syntax:

Declare a single variable:

Dim variable name [As type]

The default type is String.

Declare multiple variables. All variables will be of the specified type.

Dim variable1, variable2, ..., [As type]

Note that the shorthand syntax only applies to Dim keyword.

Declare an array:

Dim variable(Rank1, Rank2, ...) [As type]

Example: **Dim Days(7) As String**

The actual rank can be omitted for zero length arrays.

🍷 **DipToCurrent (Length As Int) As Int**

Scales the value, which represents a specific length on a default density device (Density = 1.0), to the current device.

For example, the following code will set the width value of this button to be the same physical size on all devices.

```
Button1.Width = DipToCurrent(100)
```

Note that a shorthand syntax for this method is available. Any number followed by the string 'dip' will be converted in the same manner (no spaces are allowed between the number and 'dip').

So the previous code is equivalent to:

```
Button1.Width = 100dip 'dip -> density independent pixel'
```

🍷 **DoEvents**

Processes waiting messages in the messages queue.

DoEvents can be called inside lengthy loops to allow the program to process waiting events.

🍷 **Exit**

Exits the most inner loop.

🍷 **ExitApplication**

Immediately ends the application and stops the process.

Most applications should not use this method and prefer Activity.Finish which lets the OS decide when the process is killed.

🍷 **False As Boolean**

🍷 **File As File**

Files related methods.

🍷 **Floor (Number As Double) As Double**

Returns the largest double that is smaller or equal to the specified number and is equal to an integer.

🍷 **For**

Syntax:

```
For variable = value1 To value2 [Step interval]
```

```
...
```

```
Next
```

Example:

```
For i = 1 To 10
    Log(i) 'Will print 1 to 10 (inclusive).
Next
```

If the iterator variable was not declared before it will be of type Int.

🍷 **GetDeviceLayoutValues As LayoutValues**

Returns the device LayoutValues.

Example:

```
Log(GetDeviceLayoutValues)
```

🔗GetType (object As Object) As String

Returns a string representing the object's java type.

🔗If

Single line:

If condition Then true-statement [Else false-statement]

Multiline:

If condition Then
statement

Else If condition Then
statement

...

Else
statement

End If

🔗InputList (Items As List, Title As String, CheckedItem As Int) As Int

Shows a modal dialog with a list of items and radio buttons. Pressing on an item will close the dialog.

Returns the index of the selected item or DialogResponse.Cancel if the user pressed on the back key.

List - Items to display.

Title - Dialog title.

CheckedItem - The index of the item that will first be selected. Pass -1 if no item should be preselected.

🔗InputMultiList (Items As List, Title As String) As List

Shows a modal dialog with a list of items and checkboxes. The user can select multiple items.

The dialog is closed by pressing on the "Ok" button.

Returns a list with the indices of the selected items. The list is sorted.

Returns an empty list if the user has pressed on the back key.

🔗Is

Tests whether the object is of the given type.

Example:

```
For i = 0 To Activity.NumberOfViews - 1
    If Activity.GetView(i) Is Button Then
        Dim b As Button
        b = Activity.GetView(i)
        b.Color = Colors.Blue
    End If
Next
```

🔗IsBackgroundTaskRunning (ContainerObject As Object, TaskId As Int) As Boolean

Tests whether a background task, submitted by the container object and with the specified id, is running.

🔗IsNumber (Text As String) As Boolean

Tests whether the specified string can be safely parsed as a number.

🔗 **IsPaused (Component As Object) As Boolean**

Tests whether the given component is paused. Will also return true for components that were not started yet.

Example:

```
If IsPaused(Main) = False Then CallSub(Main, "RefreshData")
```

🔗 **LastException As Exception**

Returns the last exception that was caught (if such exists).

🔗 **LoadBitmap (Dir As String, FileName As String) As Bitmap**

Loads the bitmap.

Note that the Android file system is case sensitive.

You should consider using LoadBitmapSample if the image size is large.

The actual file size is not relevant as images are usually stored compressed.

Example:

```
Activity.SetBackgroundImage(LoadBitmap(File.DirAssets, "SomeFile.jpg"))
```

🔗 **LoadBitmapSample (Dir As String, FileName As String, MaxWidth As Int, MaxHeight As Int) As Bitmap**

Loads the bitmap.

The decoder will subsample the bitmap if MaxWidth or MaxHeight are smaller than the bitmap dimensions.

This can save a lot of memory when loading large images.

Example:

```
Activity.SetBackgroundImage(LoadBitmapSample(File.DirAssets, "SomeFile.jpg",  
Activity.Width, Activity.Height))
```

🔗 **Log (Message As String)**

Logs a message. The log can be viewed in the LogCat tab.

🔗 **Logarithm (Number As Double, Base As Double) As Double**

🔗 **Max (Number1 As Double, Number2 As Double) As Double**

Returns the larger number between the two numbers.

🔗 **Me As Object**

For classes: returns a reference to the current instance.

For activities and services: returns a reference to an object that can be used with CallSub, CallSubDelayed and SubExists keywords.

Cannot be used in code modules.

🔗 **Min (Number1 As Double, Number2 As Double) As Double**

Returns the smaller number between the two numbers.

🔗Msgbox (Message As String, Title As String)

Shows a modal message box with the specified message and title.
The dialog will show one OK button.

Example:

```
Msgbox("Hello world", "This is the title")
```

🔗Msgbox2 (Message As String, Title As String, Positive As String, Cancel As String, Negative As String, Icon As android.graphics.Bitmap) As Int

Shows a modal message box with the specified message and title.

Message - The dialog message.

Title - The dialog title.

Positive - The text to show for the "positive" button. Pass "" if you don't want to show the button.

Cancel - The text to show for the "cancel" button. Pass "" if you don't want to show the button.

Negative - The text to show for the "negative" button. Pass "" if you don't want to show the button.

Icon - A bitmap that will be drawn near the title. Pass Null if you don't want to show an icon.

Returns one of the DialogResponse values.

Example:

```
Dim result As Int
result = Msgbox2("This is the message", "This is the title", "Good", "", "Bad",
LoadBitmap(File.DirAssets, "smiley.gif"))
If result = DialogResponse.Positive Then ...
```

🔗Not (Value As Boolean) As Boolean

Inverts the value of the given boolean.

🔗Null As Object

🔗NumberFormat (Number As Double, MinimumIntegers As Int, MaximumFractions As Int) As String

Converts the specified number to a string.

The string will include at least Minimum Integers and at most Maximum Fractions digits.

Example:

```
Log(NumberFormat(12345.6789, 0, 2)) '"12,345.68"'
Log(NumberFormat(1, 3, 0)) '"001"'
```

🔗NumberFormat2 (Number As Double, MinimumIntegers As Int, MaximumFractions As Int, MinimumFractions As Int, GroupingUsed As Boolean) As String

Converts the specified number to a string.

The string will include at least Minimum Integers, at most Maximum Fractions digits and at least Minimum Fractions digits.

GroupingUsed - Determines whether to group every three integers.

Example:

```
Log(NumberFormat2(12345.67, 0, 3, 3, false)) '"12345.670"'
```

🍷 **PerXToCurrent (Percentage As Float) As Int**

Returns the actual size of the given percentage of the activity width.

Example:

```
Button1.Width = PerXToCurrent(50) 'Button1.Width = 50% * Activity.Width
```

A shorthand syntax for this method is available. Any number followed by the string '%x' will be converted in the same manner (no spaces are allowed between the number and '%x'). So the previous code is equivalent to:

```
Button1.Width = 50%x
```

🍷 **PerYToCurrent (Percentage As Float) As Int**

Returns the actual size of the given percentage of the activity height.

Example:

```
Button1.Height = PerYToCurrent(50) 'Button1.Height = 50% * Activity.Height
```

A shorthand syntax for this method is available. Any number followed by the string '%y' will be converted in the same manner (no spaces are allowed between the number and '%y'). So the previous code is equivalent to:

```
Button1.Height = 50%y
```

🍷 **Power (Base As Double, Exponent As Double) As Double**

Returns the Base value raised to the Exponent power.

🍷 **ProgressDialogHide**

Hides a visible progress dialog. Does not do anything if no progress dialog is visible.

🍷 **ProgressDialogShow (Text As String)**

Shows a dialog with a circular spinning bar and the specified text.

Unlike MsgBox and InputList methods, the code will not block.

You should call ProgressDialogHide to remove the dialog.

The dialog will also be removed if the user presses on the Back key.

🍷 **ProgressDialogShow2 (Text As String, Cancelable As Boolean)**

Shows a dialog with a circular spinning bar and the specified text.

Unlike MsgBox and InputList methods, the code will not block.

You should call ProgressDialogHide to remove the dialog.

Cancelable - Whether the user can dismiss the dialog by pressing on the Back key.

🔹 **QUOTE As String**

Quote character. The value of Chr(34).

🔹 **Regex As Regex**

Regular expressions related methods.

🍷 **Return**

Returns from the current sub and optionally returns the given value.

Syntax: Return [value]

Rnd (Min As Int, Max As Int) As Int

Returns a random integer between Min (inclusive) and Max (exclusive).

RndSeed (Seed As Long)

Sets the random seed value.

This method can be used for debugging as it allows you to get the same results each time.

RndSeed (Seed As Long)

Sets the random seed value.

This method can be used for debugging as it allows you to get the same results each time.

Round (Number As Double) As Long

Returns the closest long number to the given number.

Round2 (Number As Double, DecimalPlaces As Int) As Double

Rounds the given number and leaves up to the specified number of fractional digits.

Select

Compares a single value to multiple values.

Select blocks cannot be nested.

Example:

```
Dim value As Int
value = 7
Select value
    Case 1:
        Log("One")
    Case 2, 4, 6, 8:
        Log("Even")
    Case 3, 5, 7, 9:
        Log("Odd larger than one")
    Case Else
        Log("Larger than 9")
End Select
```

Sender As Object

Returns the object that raised the event.

Only valid while inside the event sub.

Example:

```
Sub Button_Click
    Dim b As Button
    b = Sender
    b.Text = "I've been clicked"
End Sub
```

Sin (Radians As Double) As Double

Calculated the trigonometric sine function. Angle measured in radians.

SinD (Degrees As Double) As Double

Calculated the trigonometric sine function. Angle measured in degrees.

🔗 Sqrt (Value As Double) As Double

Returns the positive square root.

🔗 StartActivity (Activity As Object)

Starts an activity or brings it to front if it already exists.

Activity can be a string with the target activity name or it can be the actual activity.

After this call the current activity will be paused and the target activity will be resumed.

This method can also be used to send Intents objects to the system.

Example: StartActivity (Activity2)

🔗 StartService (Service As Object)

Starts the given service. The service will be first created if it was not started before.

The target service will be started once the program is free to process its message queue.

Note that you cannot show a MsgBox after this call and before the service starts.

Service - The service module or the service name.

Example:

StartService(SQLService)

🔗 StartServiceAt (Service As Object, Time As Long, DuringSleep As Boolean)

Schedules the given service to start at the given time.

Service - The service module or service name. Pass an empty string when calling from a service module that schedules itself.

Time - The time to start the service. If this time has already past the service will be started now.

DuringSleep - Whether to start the service when the device is sleeping. If set to false and the device is sleeping

at the specified time, the service will be started when the device wakes up.

StartServiceAt can be used to schedule a repeating task. You should call it under Service_Start to schedule the next task.

This call cancels previous scheduled tasks (for the same service).

Example:

```
StartServiceAt(SQLService, DateTime.Now + 30 * 1000, false) 'will start after 30 seconds.
```

🔗 StopService (Service As Object)

Stops the given service. Service_Destroy will be called. Call StartService afterwards will first create the service.

Service - The service module or service name. Pass empty string to stop the current service (from the service module).

Example:

StopService(SQLService)

🍷Sub

Declares a sub with the parameters and return type.

Syntax: Sub name [(list of parameters)] [As return-type]

Parameters include name and type.

The lengths of arrays dimensions should not be included.

Example:

```
Sub MySub (FirstName As String, LastName As String, Age As Int, OtherValues() As Double) As Boolean
...
End Sub
```

In this example OtherValues is a single dimension array.

The return type declaration is different than other declarations as the array parenthesis follow the type and not

the name (which does not exist in this case).

🍷SubExists (Object As Object, Sub As String) As Boolean

Tests whether the object includes the specified method.

Returns false if the object was not initialized or not an instance of a user class.

🍷TAB As String

Tab character.

🍷Tan (Radians As Double) As Double

Calculated the trigonometric tangent function. Angle measured in radians.

🍷TanD (Degrees As Double) As Double

Calculated the trigonometric tangent function. Angle measured in degrees.

🍷ToastMessageShow (Message As String, LongDuration As Boolean)

Shows a quick little message that goes out automatically.

Message - The text message to show.

LongDuration - If true then shows the message for a long period, otherwise shows the message for a short period.

🍷True As Boolean

🍷Try

Any exception thrown inside a try block will be caught in the catch block.

Call LastException to get the caught exception.

Syntax:

Try

...

Catch

...

End Try

Type

Declares a structure.

Can only be used inside sub Globals or sub Process_Globals.

Syntax:

Type type-name (field1, field2, ...)

Fields include name and type.

Example:

```
Type MyType (Name As String, Items(10) As Int)
Dim a, b As MyType
a.Initialize
a.Items(2) = 123
```

Until

Loops until the condition is true.

Syntax:

Do Until condition

...

Loop

While

Loops while the condition is true.

Syntax:

Do While condition

...

Loop

1.5 LayoutValues

Holds values related to the display.

You can get the values of the the current display by calling `GetDeviceLayoutValues`.

For example:

```
Dim lv As LayoutValues
lv = GetDeviceLayoutValues
Log(lv) 'will print the values to the log
Activity.LoadLayout and Panel.LoadLayout
return a LayoutValues object with the values of the
chosen layout variant.
```

Events:

None

Members:

 [Height](#) As Int

 [Scale](#) As Float

 [toString](#) As String

 [Width](#) As Int

Members description:

 Height As Int

The display height (pixels).

 Scale As Float

The device scale value which is equal to 'dots per inch' / 160. For most devices the value will be 1.0 or 1.5 (high resolution).

 ToString As String **Width As Int**

The display width (pixels).

1.6 String

Strings are immutable in Basic4android, which means that you can change the value of a string variable but you cannot change the text stored in a string object.

So methods like SubString, Trim and ToLowerCase return a new string, **they do not change the value of the current string**.

Typical usage:

```
Dim s As String
s = "some text"
s = s.Replace("a", "b")
```

You can use StringBuilder if you need a mutable string.

Note that string literals are also string objects:

```
Log(" some text ".Trim)
```

Events:

None

Members:

 [CharAt](#) (Index As Int) As Char

 [CompareTo](#) (Other As String) As Int

 [EndsWith](#) (Suffix As String) As Boolean

 [EqualsIgnoreCase](#) (other As String) As Boolean

 [GetBytes](#) (Charset As String) As Byte()

 [IndexOf](#) (SearchFor As String) As Int

 [IndexOf2](#) (SearchFor As String, Index As Int) As Int

 [Length](#) As Int

 [Replace](#) (Target As String, Replacement As String) As String

 [StartsWith](#) (Prefix As String) As Boolean

🔗 **Substring** (BeginIndex As Int) As String

🔗 **Substring2** (BeginIndex As Int, EndIndex As Int) As String

🔗 **ToLowerCase** As String

🔗 **ToUpperCase** As String

🔗 **Trim** As String

Members description:

🔗 **CharAt (Index As Int) As Char**

Returns the character at the given index.

🔗 **CompareTo (Other As String) As Int**

Lexicographically compares the two strings.

Returns a value less than 0 if the current string precedes Other.

Returns 0 if both strings are equal.

Returns a value larger than 0 if the current string comes after Other.

Note that upper case characters precede lower case characters.

Examples:

```
"abc".CompareTo("da") ' < 0  
"abc".CompareTo("Abc") ' > 0  
"abc".CompareTo("abca") ' < 0
```

🔗 **EndsWith (Suffix As String) As Boolean**

Returns true if this string ends with the given Suffix.

🔗 **EqualsIgnoreCase (other As String) As Boolean**

Returns true if both strings are equal ignoring their case.

🔗 **GetBytes (Charset As String) As Byte()**

Encodes the string into a new array of bytes.

Example:

```
Dim Data() As Byte  
Data = "Some string".GetBytes("UTF8")
```

🔗 **IndexOf (SearchFor As String) As Int**

Returns the index of the first occurrence of SearchFor string in the string.

Returns -1 if SearchFor was not found.

🔗 **IndexOf2 (SearchFor As String, Index As Int) As Int**

Returns the index of the first occurrence of SearchFor string in the string.

Starts searching from the given Index.

Returns -1 if SearchFor was not found.

🔗 **Length As Int**

Returns the length of this string.

🔗Replace (Target As String, Replacement As String) As String

Returns a new string resulting from the replacement of all the occurrences of Target with Replacement.

🔗StartsWith (Prefix As String) As Boolean

Returns true if this string starts with the given Prefix.

🔗SubString (BeginIndex As Int) As String

Returns a new string which is a substring of the original string.
The new string will include the character at BeginIndex and will extend to the end of the string.

Example:

```
"012345".SubString(2) 'returns "2345"
```

🔗SubString2 (BeginIndex As Int, EndIndex As Int) As String

Returns a new string which is a substring of the original string.
The new string will include the character at BeginIndex and will extend to the character at EndIndex, not including the last character.

Example:

```
"012345".SubString2(2, 4) 'returns "23"
```

🔗ToLowerCase As String

Returns a new string which is the result of lower casing this string.

🔗ToUpperCase As String

Returns a new string which is the result of upper casing this string.

🔗Trim As String

Returns a copy of the original string without any leading or trailing white spaces.

1.7 StringBuilder

StringBuilder is a mutable string, unlike regular strings which are immutable.
StringBuilder is especially useful when you need to concatenate many strings.
The following code demonstrates the performance boosting of StringBuilder:

```
Dim start As Long
start = DateTime.Now
'Regular string
Dim s As String
For i = 1 To 5000
    s = s & i
Next
Log(DateTime.Now - start)
'StringBuilder
start = DateTime.Now
Dim sb As StringBuilder
sb.Initialize
For i = 1 To 5000
    sb.Append(i)
Next
Log(DateTime.Now - start)
```

Tested on a real device, the first 'for loop' took about 20 seconds and the second took less then tenth of a second.

The reason is that the code: `s = s & i` creates a new string each iteration (strings are immutable).

The method `StringBuilder.ToString` converts the object to a string.

Events:

None

Members:

 [Append](#) (Text As String) As StringBuilder

 [Initialize](#)

 [Insert](#) (Offset As Int, Text As String) As StringBuilder

 [IsInitialized](#) As Boolean

 [Length](#) As Int [read only]

 [Remove](#) (StartOffset As Int, EndOffset As Int) As StringBuilder

 [ToString](#) As String

Members description:

 **Append** (Text As String) As StringBuilder

Appends the specified text at the end.

Returns the same object, so you can chain methods.

Example:

```
sb.Append("First line").Append(CRLF).Append("Second line")
```

 **Initialize**

Initializes the object.

Example:

```
Dim sb As StringBuilder
```

```
sb.Initialize
```

```
sb.Append("The value is: ").Append(SomeOtherVariable).Append(CRLF)
```

 **Insert** (Offset As Int, Text As String) As StringBuilder

Inserts the specified text at the specified offset.

 **IsInitialized** As Boolean

 **Length** As Int [read only]

Returns the number of characters.

🔧 Remove (StartOffset As Int, EndOffset As Int) As StringBuilder

Removes the specified characters.

StartOffset - The first character to remove.

EndOffset - The ending index. This character will not be removed.

🔧 ToString As String

Converts the object to a string.

1.8 Timer

A Timer object generates ticks events at specified intervals.

Using a timer is a good alternative to a long loop, as it allows the UI thread to handle other events and messages.

Note that the timer events will not fire while the UI thread is busy running other code (unless you call DoEvents keyword).

The timer Enabled property is set to False by default. To make it start working you should change it to True.

Timer events will not fire when the activity is paused, or if a blocking dialog (like MsgBox) is visible.

Timers should be declared in Sub Process_Globals. Otherwise you may get multiple timers running when the activity is recreated.

It is also important to disable the timer when the activity is pausing and then enable it when it resumes. This will save CPU and battery.

Events:

Tick

Members:



Enabled As Boolean



Initialize (EventName As String, Interval As Long)



Interval As Long

Members description:



Enabled As Boolean

Gets or sets whether the timer is enabled (ticking).



Initialize (EventName As String, Interval As Long)

Initializes the timer with the event sub prefix and the specified interval (measured in milliseconds).
IMPORTANT: this object should be declared in Sub Process_Globals.

Example:

```
Timer1.Initialize("Timer1", 1000)
Timer1.Enabled = True
```

```
Sub Timer1_Tick
'Handle tick events
End Sub
```



Interval As Long

Gets or sets the interval between tick events, measured in milliseconds.

2 Views

List of types:

[Activity](#)
[Button](#)
[CheckBox](#)
[EditText](#)
[ImageView](#)
[Label](#)
[ListView](#)
[Panel](#)
[ProgressBar](#)
[RadioButton](#)
[ScrollView](#)
[SeekBar](#)
[Spinner](#)
[TabHost](#)
[ToggleButton](#)
[View](#)
[WebView](#)

2.1 Activity

Each activity module include a predefined Activity object.

Activity is the main component of your application.

Activities have three special life cycle related event: Activity_Create, Activity_Resume and Activity_Pause.

See this tutorial for more information about activities and processes life cycle: [Life cycle tutorial](#).

You can add and remove views to this activity with AddView and RemoveViewAt methods.

You can also load a layout file with LoadLayout.

The Touch event can be used to handle user touches.

The first parameter of this event is the Action parameter. The parameter values can be ACTION_DOWN,

ACTION_MOVE or ACTION_UP. Use this value to find the user current action.

The KeyPress event occurs when the user has pressed on the key and assuming that no other view has consumed this event (like EditText).

When handling the KeyPress event you should return a boolean value which tells whether the key press event was consumed.

For example if the user pressed on the Back key and you return True then the OS will not close your activity.

```
Sub Activity_KeyPress (KeyCode As Int) As Boolean
    If KeyCode = KeyCodes.KEYCODE_BACK Then
        Return True
    Else
        Return False
    End If
End Sub
```

You can add menu items to the activity with AddMenuItem method. Note that this method should

only be called inside
Activity_Create event.

Events:

Touch (Action As Int, X As Float, Y As Float)

KeyPress (KeyCode As Int) As Boolean 'return true if you want to consume the event

Click

LongClick

Members:

 [ACTION_DOWN](#) As Int

 [ACTION_MOVE](#) As Int

 [ACTION_UP](#) As Int

 [AddMenuItem](#) (Title As String, EventName As String)

 [AddMenuItem2](#) (Title As String, EventName As String, Bitmap As android.graphics.Bitmap)

 [AddView](#) (View As android.view.View, Left As Int, Top As Int, Width As Int, Height As Int)

 [Background](#) As android.graphics.drawable.Drawable

 [BringToFront](#)

 [Color](#) As Int [write only]

 [Finish](#)

 [GetView](#) (Index As Int) As View

 [Height](#) As Int

 [Initialize](#) (EventName As String)

 [Invalidate](#)

 [Invalidate2](#) (Rect As android.graphics.Rect)

 [Invalidate3](#) (Left As Int, Top As Int, Right As Int, Bottom As Int)

 [IsInitialized](#) As Boolean

 [Left](#) As Int

 [LoadLayout](#) (Layout As String) As LayoutValues

 [NumberOfViews](#) As Int [read only]

 [RemoveView](#)

 **RemoveViewAt** (Index **As Int**)

 **RequestFocus** **As Boolean**

 **SendToBack**

 **SetBackgroundImage** (**Bitmap As** android.graphics.Bitmap)

 **SetLayout** (Left **As Int**, Top **As Int**, Width **As Int**, Height **As Int**)

 **Tag** **As Object**

 **Title** **As CharSequence**

 **Top** **As Int**

 **Width** **As Int**

Members description:

 **ACTION_DOWN** **As Int**

 **ACTION_MOVE** **As Int**

 **ACTION_UP** **As Int**

 **AddMenuItem** (Title **As String**, EventName **As String**)

Adds a menu item to the activity.

Title - Menu item title.

EventName - The prefix name of the sub that will handle the click event.

This method should only be called inside sub Activity_Create.

Note that the 'Sender' value inside the click event equals to the clicked menu item text.

Example:

```
Activity.AddMenuItem("Open File", "OpenFile")
```

```
...
```

```
Sub OpenFile_Click
```

```
...
```

```
End Sub
```

 **AddMenuItem2** (Title **As String**, EventName **As String**, **Bitmap As** android.graphics.Bitmap)

Adds a menu item to the activity.

Title - Menu item title.

EventName - The prefix name of the sub that will handle the click event.

Bitmap - Bitmap to draw as the item background.

Only the first five (or six if there are six total) menu items display icons.

This method should only be called inside sub Activity_Create.

Note that the 'Sender' value inside the click event equals to the clicked menu item text.

Example:

```
Activity.AddMenuItem2("Open File", "OpenFile",  
LoadBitmap(File.DirAssets, "SomeImage.png"))
```

```
...  
Sub OpenFile_Click  
...  
End Sub
```

 **AddView** (**View** As android.view.View, Left As Int, Top As Int, Width As Int, Height As Int)

Adds a view to this activity.

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Finish**

Closes this activity.

 **GetView** (Index As Int) As View

Gets the view that is stored in the specified index.

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized** As Boolean

 **Left** As Int

🔧📄 **LoadLayout (Layout As String) As LayoutValues**

Loads a layout file (.bal).

Returns the LayoutValues of the actual layout variant that was loaded.

🔧📄 **NumberOfViews As Int [read only]**

Returns the number of child views.

🔧📄 **RemoveView**

Removes this view from its parent.

🔧📄 **RemoveViewAt (Index As Int)**

Removes the view that is stored in the specified index.

🔧📄 **RequestFocus As Boolean**

Tries to set the focus to this view.

Returns True if the focus was set.

🔧📄 **SendToBack**

Changes the Z order of this view and sends it to the back.

🔧📄 **SetBackgroundImage (Bitmap As android.graphics.Bitmap)**

🔧📄 **SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)**

Changes the view position and size.

🔧📄 **Tag As Object**

Gets or sets the Tag value. This is a place holder which can used to store additional data.

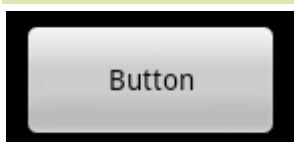
🔧📄 **Title As CharSequence**

🔧📄 **Top As Int**

🔧📄 **Width As Int**

Gets or sets the view's width.

2.2 Button



A Button view.

If you change the button's background you will usually want to use StateListDrawable which allows you to set the "default" drawable and the "pressed" drawable.

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

Down

Up

Click

LongClick

Members:

 **Background** As android.graphics.drawable.Drawable

 **BringToFront**

 **Color** As Int [write only]

 **Enabled** As Boolean

 **Gravity** As Int

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

 **Invalidate2** (Rect As android.graphics.Rect)

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

 **IsInitialized** As Boolean

 **Left** As Int

 **RemoveView**

 **RequestFocus** As Boolean

 **SendToBack**

 **SetBackgroundImage** (Bitmap As android.graphics.Bitmap)

 **SetLayout** (Left As Int, Top As Int, Width As Int, Height As Int)

 **Tag** As Object

 **Text** As String

 **TextColor** As Int

 **TextSize** As Float

 **Top** As Int

 **Typeface** As android.graphics.Typeface

 **Visible** As Boolean

 **Width** As Int

Members description:

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean

 **Gravity** As Int

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized** As Boolean

 **Left** As Int

 **RemoveView**

Removes this view from its parent.

 **RequestFocus** As Boolean

Tries to set the focus to this view.
Returns True if the focus was set.

SendToBack

Changes the Z order of this view and sends it to the back.

SetBackgroundImage (**Bitmap** As **android.graphics.Bitmap**)

SetLayout (**Left** As **Int**, **Top** As **Int**, **Width** As **Int**, **Height** As **Int**)

Changes the view position and size.

Tag As **Object**

Gets or sets the Tag value. This is a place holder which can used to store additional data.

Text As **String**

TextColor As **Int**

TextSize As **Float**

Top As **Int**

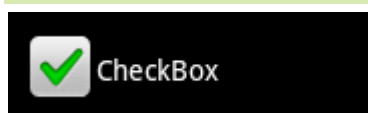
Typeface As **android.graphics.Typeface**

Visible As **Boolean**

Width As **Int**

Gets or sets the view's width.

2.3 CheckBox



A CheckBox view. Unlike RadioButtons each CheckBox can be checked independently. This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

CheckedChange(Checked As Boolean)

Members:

 **Background** As **android.graphics.drawable.Drawable**

 **BringToFront**

 **Checked** As **Boolean**

 **Color** As **Int** [write only]

 **Enabled** As Boolean

 **Gravity** As Int

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

 **Invalidate2** (Rect As android.graphics.Rect)

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

 **IsInitialized** As Boolean

 **Left** As Int

 **RemoveView**

 **RequestFocus** As Boolean

 **SendToBack**

 **SetBackgroundImage** (Bitmap As android.graphics.Bitmap)

 **SetLayout** (Left As Int, Top As Int, Width As Int, Height As Int)

 **Tag** As Object

 **Text** As String

 **TextColor** As Int

 **TextSize** As Float

 **Top** As Int

 **Typeface** As android.graphics.Typeface

 **Visible** As Boolean

 **Width** As Int

Members description:

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 Checked As Boolean **Color As Int [write only]**

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 Enabled As Boolean **Gravity As Int** **Height As Int** **Initialize (EventName As String)** **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 Invalidate2 (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 Invalidate3 (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 IsInitialized As Boolean **Left As Int** **RemoveView**

Removes this view from its parent.

 RequestFocus As Boolean

Tries to set the focus to this view.

Returns True if the focus was set.

 SendToBack

Changes the Z order of this view and sends it to the back.

 SetBackgroundImage (Bitmap As android.graphics.Bitmap) **SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)**

Changes the view position and size.

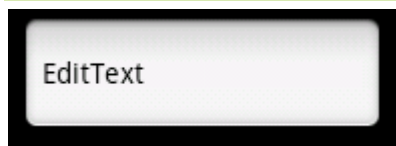
 Tag As Object

Gets or sets the Tag value. This is a place holder which can used to store additional data.

 Text As String **TextColor As Int** **TextSize As Float** **Top As Int** **Typeface As android.graphics.Typeface** **Visible As Boolean** **Width As Int**

Gets or sets the view's width.

2.4 EditText



EditText view is a view that allows the user to write free text (similar to WinForms TextBox).

The EditText has two modes; SingleLine and MultiLine. You can set it to be multiline by calling `EditText1.SingleLine = False`

On most devices the soft keyboard will show automatically when the user presses on the EditText. You can change the InputType property and change the type of keyboard that appears.

For example: `EditText1.InputType = EditText1.INPUT_TYPE_NUMBERS` will cause the numeric keyboard to appear when the user presses on the EditText. Note that it will also cause the EditText to only accept numbers.

The TextChanged event fires whenever the text changes and it includes the old and new strings.

The EnterPressed event fires when the user presses on the enter key or action key (Done or Next).

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

TextChanged (Old As String, New As String)

EnterPressed

Members:

 **Background As android.graphics.drawable.Drawable** **BringToFront** **Color As Int** [write only] **Enabled As Boolean**

-  **ForceDoneButton** As Boolean [write only]
-  **Gravity** As Int
-  **Height** As Int
-  **Hint** As String
-  **HintColor** As Int
-  **Initialize** (EventName As String)
-  **INPUT_TYPE_DECIMAL_NUMBERS** As Int
-  **INPUT_TYPE_NONE** As Int
-  **INPUT_TYPE_NUMBERS** As Int
-  **INPUT_TYPE_PHONE** As Int
-  **INPUT_TYPE_TEXT** As Int
-  **InputType** As Int
-  **Invalidate**
-  **Invalidate2** (Rect As android.graphics.Rect)
-  **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)
-  **IsInitialized** As Boolean
-  **Left** As Int
-  **PasswordMode** As Boolean [write only]
-  **RemoveView**
-  **RequestFocus** As Boolean
-  **SelectAll**
-  **SelectionStart** As Int
-  **SendToBack**
-  **SetBackgroundImage** (Bitmap As android.graphics.Bitmap)
-  **SetLayout** (Left As Int, Top As Int, Width As Int, Height As Int)
-  **SingleLine** As Boolean [write only]

 **Tag** As Object

 **Text** As String

 **TextColor** As Int

 **TextSize** As Float

 **Top** As Int

 **Typeface** As android.graphics.Typeface

 **Visible** As Boolean

 **Width** As Int

 **Wrap** As Boolean [write only]

Members description:

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean

 **ForceDoneButton** As Boolean [write only]

By default the OS sets the virtual keyboard action key to display Done or Next according to the specific layout.

You can force it to display Done by setting this value to True.

Example:

```
EditText1.ForceDoneButton = True
```

 **Gravity** As Int

 **Height** As Int

 **Hint** As String

Gets or sets the text that will appear when the EditText is empty.

Example:

```
EditText1.Hint = "Enter username"
```

 **HintColor** As Int

Gets or sets the hint text color.

Example:

```
EditText1.HintColor = Colors.Gray
```

🔧 **Initialize (EventName As String)**

🔧 **INPUT_TYPE_DECIMAL_NUMBERS As Int**

Numeric keyboard will be displayed. Numbers, decimal point and minus sign are accepted.

🔧 **INPUT_TYPE_NONE As Int**

No keyboard will be displayed.

🔧 **INPUT_TYPE_NUMBERS As Int**

Numeric keyboard will be displayed. Only numbers are accepted.

🔧 **INPUT_TYPE_PHONE As Int**

Keyboard will be displayed in phone mode.

🔧 **INPUT_TYPE_TEXT As Int**

Default text mode.

🔧 **InputType As Int**

Gets or sets the input type flag. This flag is used to determine the settings of the virtual keyboard.

Example:

```
EditText1.InputType = EditText1.INPUT_TYPE_NUMBERS
```

🔧 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

🔧 **Invalidate2 (Rect As android.graphics.Rect)**

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

🔧 **Invalidate3 (Left As Int, Top As Int, Right As Int, Bottom As Int)**

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

🔧 **IsInitialized As Boolean**

🔧 **Left As Int**

🔧 **PasswordMode As Boolean [write only]**

Sets whether the EditText should be in password mode and hide the actual characters.

 RemoveView

Removes this view from its parent.

 RequestFocus **As Boolean**

Tries to set the focus to this view.
Returns True if the focus was set.

 SelectAll

Selects the entire text.

 SelectionStart **As Int**

Gets or sets the selection start position (or the cursor position).
Returns -1 if there is no selection or cursor.

 SendToBack

Changes the Z order of this view and sends it to the back.

 SetBackgroundImage (**Bitmap** **As** **android.graphics.Bitmap**) **SetLayout** (**Left** **As Int**, **Top** **As Int**, **Width** **As Int**, **Height** **As Int**)

Changes the view position and size.

 SingleLine **As Boolean** [write only]

Sets whether the EditText should be in single line mode or multiline mode.

 Tag **As Object**

Gets or sets the Tag value. This is a place holder which can be used to store additional data.

 Text **As String** **TextColor** **As Int** **TextSize** **As Float** **Top** **As Int** **Typeface** **As** **android.graphics.Typeface** **Visible** **As Boolean** **Width** **As Int**

Gets or sets the view's width.

 Wrap **As Boolean** [write only]

Sets whether the text content will wrap within the EditText bounds. Relevant when the EditText is in multiline mode.

Example:

```
EditText1.Wrap = False
```

2.5 ImageView



A view that shows an image.

You can assign a bitmap using the Bitmap property.

The Gravity property changes the way the image appears.

The two most relevant values are Gravity.FILL (which will cause the image to fill the entire view) and Gravity.CENTER (which will draw the image in the view's center).

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

Click

LongClick

Members:

 **Background** As android.graphics.drawable.Drawable

 **Bitmap** As android.graphics.Bitmap

 **BringToFront**

 **Color** As Int [write only]

 **Enabled** As Boolean

 **Gravity** As Int

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

 **Invalidate2** (Rect As android.graphics.Rect)

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

 **IsInitialized** As Boolean

 **Left** As Int

 [RemoveView](#)

 [RequestFocus](#) As Boolean

 [SendToBack](#)

 [SetBackgroundImage](#) (Bitmap As android.graphics.Bitmap)

 [SetLayout](#) (Left As Int, Top As Int, Width As Int, Height As Int)

 [Tag](#) As Object

 [Top](#) As Int

 [Visible](#) As Boolean

 [Width](#) As Int

Members description:

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **Bitmap** As android.graphics.Bitmap

Gets or sets the bitmap assigned to the ImageView.

Example:

```
ImageView1.Bitmap = LoadBitmap(File.DirAssets, "someimage.jpg")
```

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean

 **Gravity** As Int

Gets or sets the gravity assigned to the bitmap.

Example:

```
ImageView1.Gravity = Gravity.Fill
```

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

Invalidate2 (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

Invalidate3 (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

IsInitialized As Boolean

Left As Int

RemoveView

Removes this view from its parent.

RequestFocus As Boolean

Tries to set the focus to this view.

Returns True if the focus was set.

SendToBack

Changes the Z order of this view and sends it to the back.

SetBackgroundImage (Bitmap As android.graphics.Bitmap)

SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)

Changes the view position and size.

Tag As Object

Gets or sets the Tag value. This is a place holder which can used to store additional data.

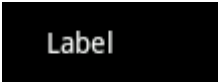
Top As Int

Visible As Boolean

Width As Int

Gets or sets the view's width.

2.6 Label



A Label view that shows read-only text.

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

Click

LongClick

Members:

 [Background](#) As android.graphics.drawable.Drawable

 [BringToFront](#)

 [Color](#) As Int [write only]

 [Enabled](#) As Boolean

 [Gravity](#) As Int

 [Height](#) As Int

 [Initialize](#) (EventName As String)

 [Invalidate](#)

 [Invalidate2](#) (Rect As android.graphics.Rect)

 [Invalidate3](#) (Left As Int, Top As Int, Right As Int, Bottom As Int)

 [IsInitialized](#) As Boolean

 [Left](#) As Int

 [RemoveView](#)

 [RequestFocus](#) As Boolean

 [SendToBack](#)

 [SetBackgroundImage](#) (Bitmap As android.graphics.Bitmap)

 [SetLayout](#) (Left As Int, Top As Int, Width As Int, Height As Int)

 [Tag](#) As Object

 [Text](#) As String

 **TextColor** As Int

 **TextSize** As Float

 **Top** As Int

 **Typeface** As android.graphics.Typeface

 **Visible** As Boolean

 **Width** As Int

Members description:

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean

 **Gravity** As Int

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized** As Boolean

 **Left As Int**

 **RemoveView**

Removes this view from its parent.

 **RequestFocus As Boolean**

Tries to set the focus to this view.
Returns True if the focus was set.

 **SendToBack**

Changes the Z order of this view and sends it to the back.

 **SetBackgroundImage (Bitmap As android.graphics.Bitmap)**

 **SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)**

Changes the view position and size.

 **Tag As Object**

Gets or sets the Tag value. This is a place holder which can used to store additional data.

 **Text As String**

 **TextColor As Int**

 **TextSize As Float**

 **Top As Int**

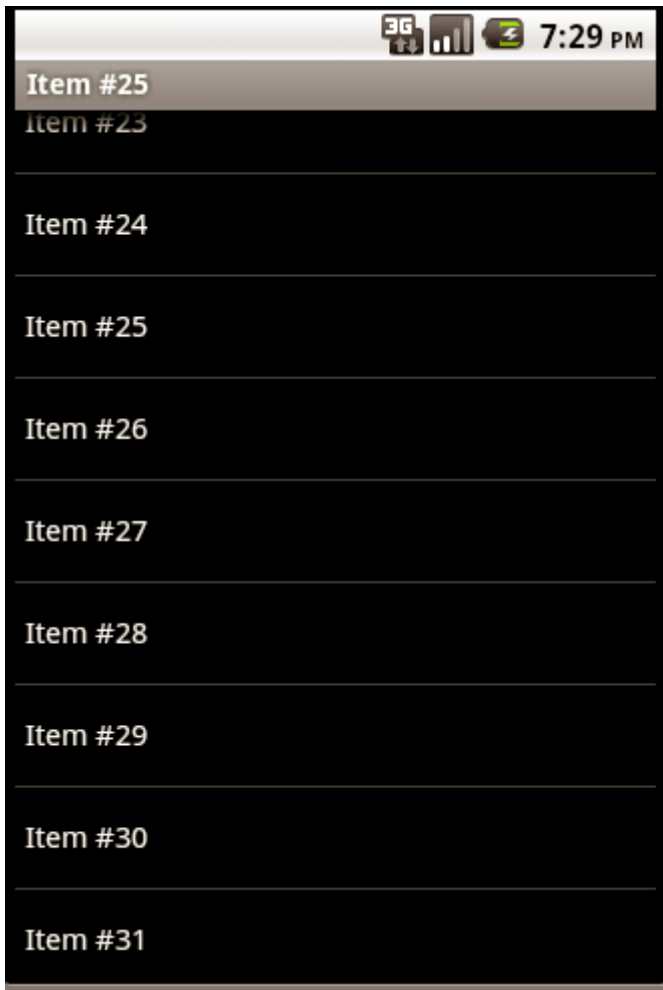
 **Typeface As android.graphics.Typeface**

 **Visible As Boolean**

 **Width As Int**

Gets or sets the view's width.

2.7 ListView



ListView is a very useful view that can handle large and small lists.

The ListView raises two events. ItemClick is raised when an item is clicked and ItemLongClick is raised when an item is clicked and held.

See the [ListView tutorial](#) for more information.

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

ItemClick (Position As Int, Value As Object)

ItemLongClick (Position As Int, Value As Object)

Members:

🔧 [AddSingleLine](#) (Text As String)

🔧 [AddSingleLine2](#) (Text As String, ReturnValue As Object)

🔧 [AddTwoLines](#) (Text1 As String, Text2 As String)

🔧 [AddTwoLines2](#) (Text1 As String, Text2 As String, ReturnValue As Object)

🔧 [AddTwoLinesAndBitmap](#) (Text1 As String, Text2 As String, Bitmap As android.graphics.Bitmap)

 **AddTwoLinesAndBitmap2** (Text1 *As String*, Text2 *As String*, Bitmap *As android.graphics.Bitmap*, ReturnValue *As Object*)

 **Background** *As android.graphics.drawable.Drawable*

 **BringToFront**

 **Clear**

 **Color** *As Int* [write only]

 **Enabled** *As Boolean*

 **FastScrollEnabled** *As Boolean*

 **GetItem** (Index *As Int*) *As Object*

 **Height** *As Int*

 **Initialize** (EventName *As String*)

 **Invalidate**

 **Invalidate2** (Rect *As android.graphics.Rect*)

 **Invalidate3** (Left *As Int*, Top *As Int*, Right *As Int*, Bottom *As Int*)

 **IsInitialized** *As Boolean*

 **Left** *As Int*

 **RemoveAt** (Index *As Int*)

 **RemoveView**

 **RequestFocus** *As Boolean*

 **ScrollingBackgroundColor** *As Int* [write only]

 **SendToBack**

 **SetBackgroundImage** (Bitmap *As android.graphics.Bitmap*)

 **SetLayout** (Left *As Int*, Top *As Int*, Width *As Int*, Height *As Int*)

 **SingleLineLayout** *As SingleLineLayout* [read only]

 **Size** *As Int* [read only]

 **Tag** *As Object*

 [Top](#) As Int

 [TwoLinesAndBitmap](#) As TwoLinesAndBitmapLayout [read only]

 [TwoLinesLayout](#) As TwoLinesLayout [read only]

 [Visible](#) As Boolean

 [Width](#) As Int

Members description:

 **AddSingleLine** (Text As String)

Adds a single line item.

Example:

```
ListView1.AddSingleLine("Sunday")
```

 **AddSingleLine2** (Text As String, ReturnValue As Object)

Adds a single line item.

The specified return value will be returned when calling GetItem or in the ItemClick event.

Example:

```
ListView1.AddSingleLine2("Sunday", 1)
```

 **AddTwoLines** (Text1 As String, Text2 As String)

Adds a two lines item.

Example:

```
ListView1.AddTwoLines("This is the first line.", "And this is the  
second")
```

 **AddTwoLines2** (Text1 As String, Text2 As String, ReturnValue As Object)

Adds a two lines item.

The specified return value will be returned when calling GetItem or in the ItemClick event.

 **AddTwoLinesAndBitmap** (Text1 As String, Text2 As String, Bitmap As android.graphics.Bitmap)

Adds a two lines and a bitmap item.

Example:

```
ListView1.AddTwoLinesAndBitmap("First line", "Second line",  
LoadBitmap(File.DirAssets, "SomeImage.png"))
```

 **AddTwoLinesAndBitmap2** (Text1 As String, Text2 As String, Bitmap As android.graphics.Bitmap, ReturnValue As Object)

Adds a two lines and a bitmap item.

The specified return value will be returned when calling GetItem or in the ItemClick event.

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

Clear

Clears all items from the list.

Color As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

Enabled As Boolean

FastScrollEnabled As Boolean

Gets or sets whether the fast scroll icon will appear when the user scrolls the list.
The default is false.

GetItem (Index As Int) As Object

Returns the value of the item at the specified position.

Returns the "return value" if it was set and if not returns the text of the first line.

Height As Int

Initialize (EventName As String)

Invalidate

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

Invalidate2 (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

Invalidate3 (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

IsInitialized As Boolean

Left As Int

RemoveAt (Index As Int)

Removes the item at the specified position.

RemoveView

Removes this view from its parent.

RequestFocus As Boolean

Tries to set the focus to this view.
Returns True if the focus was set.

ScrollingBackgroundColor As Int [write only]

Sets the background color that will be used while scrolling the list.
This is an optimization done to make the scrolling smoother.
Set to `Colors.Transparent` if the background behind the list is not solid color.
The default is black.

SendToBack

Changes the Z order of this view and sends it to the back.

SetBackgroundImage (Bitmap As android.graphics.Bitmap)

SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)

Changes the view position and size.

SingleLineLayout As SingleLineLayout [read only]

Returns the layout that is used to show single line items.
You can change the layout values to change the appearance of such items.
Example:

```
Dim Label1 As Label
Label1 = ListView1.SingleLineLayout.Label
Label1.TextSize = 20
Label1.TextColor = Colors.Green
```

Size As Int [read only]

Returns the number of items stored in the list.

Tag As Object

Gets or sets the Tag value. This is a place holder which can used to store additional data.

Top As Int

TwoLinesAndBitmap As TwoLinesAndBitmapLayout [read only]

Returns the layout that is used to show two lines and bitmap items.
You can change the layout values to change the appearance of such items.
For example if you want to remove the second label (in all items with this layout):
`ListView1.TwoLinesAndBitmap.SecondLabel.Visible = False`

TwoLinesLayout As TwoLinesLayout [read only]

Returns the layout that is used to show two lines items.
You can change the layout values to change the appearance of such items.
Example:

```
Dim Label1 As Label
Label1 = ListView1.TwoLinesLayout.SecondLabel
```

```
Label1.TextSize = 20  
Label1.TextColor = Colors.Green
```

 **Visible** As Boolean

 **Width** As Int

Gets or sets the view's width.

2.8 Panel



A Panel is a view that holds other child views.

You can add child views programmatically or by loading a layout file.

The Panel raises the Touch event. The first parameter of this event is the Action which is one of the Activity action constants.

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

Touch (Action As Int, X As Float, Y As Float)

Click

LongClick

Members:

 **AddView** (View As android.view.View, Left As Int, Top As Int, Width As Int, Height As Int)

 **Background** As android.graphics.drawable.Drawable

 **BringToFront**

 **Color** As Int [write only]

 **Enabled** As Boolean

 **GetView** (Index As Int) As View

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

-  **Invalidate2** (Rect As android.graphics.Rect)
-  **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)
-  **IsInitialized** As Boolean
-  **Left** As Int
-  **LoadLayout** (Layout As String) As LayoutValues
-  **NumberOfViews** As Int [read only]
-  **RemoveView**
-  **RemoveViewAt** (Index As Int)
-  **RequestFocus** As Boolean
-  **SendToBack**
-  **SetBackgroundImage** (Bitmap As android.graphics.Bitmap)
-  **SetLayout** (Left As Int, Top As Int, Width As Int, Height As Int)
-  **Tag** As Object
-  **Top** As Int
-  **Visible** As Boolean
-  **Width** As Int

Members description:

 **AddView** (View As android.view.View, Left As Int, Top As Int, Width As Int, Height As Int)

Adds a view to this panel.

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean

 GetView (Index As Int) As View

Gets the view that is stored in the specified index.

 Height As Int **Initialize (EventName As String)** **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 Invalidate2 (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 Invalidate3 (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 IsInitialized As Boolean **Left As Int** **LoadLayout (Layout As String) As LayoutValues**

Loads a layout file to this panel. Returns the value of the chosen layout variant.

 NumberOfViews As Int [read only]

Returns the number of child views.

 RemoveView

Removes this view from its parent.

 RemoveViewAt (Index As Int)

Removes the view that is stored in the specified index.

 RequestFocus As Boolean

Tries to set the focus to this view.

Returns True if the focus was set.

 SendToBack

Changes the Z order of this view and sends it to the back.

 SetBackgroundImage (Bitmap As android.graphics.Bitmap) **SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)**

Changes the view position and size.

Tag As Object

Gets or sets the Tag value. This is a place holder which can used to store additional data.

Top As Int

Visible As Boolean

Width As Int

Gets or sets the view's width.

2.9 ProgressBar



A progress bar view. The Progress property sets the progress value which is between 0 to 100. This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

None

Members:

 **Background** As android.graphics.drawable.Drawable

 **BringToFront**

 **Color** As Int [write only]

 **Enabled** As Boolean

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

 **Invalidate2** (Rect As android.graphics.Rect)

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

 **IsInitialized** As Boolean

 **Left** As Int

 **Progress** As Int

 **RemoveView**

 **RequestFocus** As Boolean

 **SendToBack**

 **SetBackgroundImage** (Bitmap As android.graphics.Bitmap)

 **SetLayout** (Left As Int, Top As Int, Width As Int, Height As Int)

 **Tag** As Object

 **Top** As Int

 **Visible** As Boolean

 **Width** As Int

Members description:

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized** *As Boolean*

 **Left** *As Int*

 **Progress** *As Int*

Gets or sets the progress value.

 **RemoveView**

Removes this view from its parent.

 **RequestFocus** *As Boolean*

Tries to set the focus to this view.

Returns True if the focus was set.

 **SendToBack**

Changes the Z order of this view and sends it to the back.

 **SetBackgroundImage** (*Bitmap* *As* *android.graphics.Bitmap*)

 **SetLayout** (*Left* *As Int*, *Top* *As Int*, *Width* *As Int*, *Height* *As Int*)

Changes the view position and size.

 **Tag** *As Object*

Gets or sets the Tag value. This is a place holder which can used to store additional data.

 **Top** *As Int*

 **Visible** *As Boolean*

 **Width** *As Int*

Gets or sets the view's width.

2.10 RadioButton



A RadioButton view. Only one RadioButton in a group can be checked. When a different RadioButton is checked all others will automatically be unchecked. Grouping is done by adding RadioButtons to the same activity or panel.

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

CheckedChange(Checked As Boolean)

Members:

-  **Background** As android.graphics.drawable.Drawable
-  **BringToFront**
-  **Checked** As Boolean
-  **Color** As Int [write only]
-  **Enabled** As Boolean
-  **Gravity** As Int
-  **Height** As Int
-  **Initialize** (EventName As String)
-  **Invalidate**
-  **Invalidate2** (Rect As android.graphics.Rect)
-  **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)
-  **IsInitialized** As Boolean
-  **Left** As Int
-  **RemoveView**
-  **RequestFocus** As Boolean
-  **SendToBack**
-  **SetBackgroundImage** (Bitmap As android.graphics.Bitmap)
-  **SetLayout** (Left As Int, Top As Int, Width As Int, Height As Int)
-  **Tag** As Object
-  **Text** As String
-  **TextColor** As Int
-  **TextSize** As Float
-  **Top** As Int
-  **Typeface** As android.graphics.Typeface

 **Visible** As Boolean

 **Width** As Int

Members description:

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Checked** As Boolean

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean

 **Gravity** As Int

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized** As Boolean

 **Left** As Int

 **RemoveView**

Removes this view from its parent.

 **RequestFocus** As Boolean

Tries to set the focus to this view.
Returns True if the focus was set.

📌 **SendToBack**

Changes the Z order of this view and sends it to the back.

📌 **SetBackgroundImage** (**Bitmap As** android.graphics.Bitmap)

📌 **SetLayout** (**Left As Int**, **Top As Int**, **Width As Int**, **Height As Int**)

Changes the view position and size.

📌 **Tag As Object**

Gets or sets the Tag value. This is a place holder which can used to store additional data.

📌 **Text As String**

📌 **TextColor As Int**

📌 **TextSize As Float**

📌 **Top As Int**

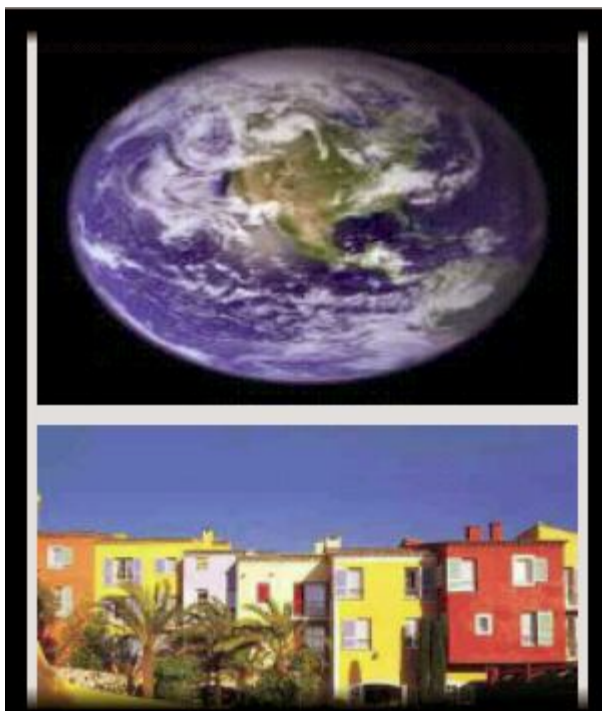
📌 **Typeface As** android.graphics.Typeface

📌 **Visible As Boolean**

📌 **Width As Int**

Gets or sets the view's width.

2.11 ScrollView



ScrollView is a view that contains other views and allows the user to vertically scroll those views. See the [ScrollView tutorial](#) for more information.

The ScrollView has an inner panel which actually contains the child views.

You can add views by calling: `ScrollView1.Panel.AddView(...)`

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

None

Members:

 [Background](#) As android.graphics.drawable.Drawable

 [BringToFront](#)

 [Color](#) As Int [write only]

 [Enabled](#) As Boolean

 [FullScroll](#) (Bottom As Boolean)

 [Height](#) As Int

 [Initialize](#) (Height As Int)

 [Invalidate](#)

 [Invalidate2](#) (Rect As android.graphics.Rect)

 [Invalidate3](#) (Left As Int, Top As Int, Right As Int, Bottom As Int)

 [IsInitialized](#) As Boolean

 [Left](#) As Int

 [Panel](#) As Panel [read only]

 [RemoveView](#)

 [RequestFocus](#) As Boolean

 [ScrollPosition](#) As Int

 [SendToBack](#)

 [SetBackgroundImage](#) (Bitmap As android.graphics.Bitmap)

 [SetLayout](#) (Left As Int, Top As Int, Width As Int, Height As Int)

 [Tag](#) As Object

 **Top** As Int **Visible** As Boolean **Width** As Int

Members description:

 **Background** As **android.graphics.drawable.Drawable**

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean **FullScroll** (**Bottom** As Boolean)

Scrolls the scroll view to the top or bottom.

 **Height** As Int **Initialize** (**Height** As Int)

Initializes the ScrollView and sets its inner panel height to the given height.

You can later change this height by calling ScrollView.Panel.Height.

```
Dim ScrollView1 As ScrollView
ScrollView1.Initialize(1000dip)
```

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (**Rect** As **android.graphics.Rect**)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3** (**Left** As Int, **Top** As Int, **Right** As Int, **Bottom** As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized** As Boolean

 Left As Int **Panel As Panel [read only]**

Returns the panel which you can use to add views to.

Example:

```
ScrollView1.Panel.AddView(...)
```

 RemoveView

Removes this view from its parent.

 RequestFocus As Boolean

Tries to set the focus to this view.

Returns True if the focus was set.

 ScrollPosition As Int

Gets or sets the scroll position.

 SendToBack

Changes the Z order of this view and sends it to the back.

 SetBackgroundImage (Bitmap As android.graphics.Bitmap) **SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)**

Changes the view position and size.

 Tag As Object

Gets or sets the Tag value. This is a place holder which can used to store additional data.

 Top As Int **Visible As Boolean** **Width As Int**

Gets or sets the view's width.

2.12 SeekBar



A view that allows the user to set a value by dragging a slider. Similar to WinForms TrackBar. The ValueChanged event is raised whenever the value is changed. The UserChanged parameter can be used to distinguish between changes done by the user and changes done programmatically. This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

ValueChanged (Value As Int, UserChanged As Boolean)

Members:

 **Background** As android.graphics.drawable.Drawable

 **BringToFront**

 **Color** As Int [write only]

 **Enabled** As Boolean

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

 **Invalidate2** (Rect As android.graphics.Rect)

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

 **IsInitialized** As Boolean

 **Left** As Int

 **Max** As Int

 **RemoveView**

 **RequestFocus** As Boolean

 **SendToBack**

 **SetBackgroundImage** (Bitmap As android.graphics.Bitmap)

 **SetLayout** (Left As Int, Top As Int, Width As Int, Height As Int)

 **Tag** As Object

 **Top** As Int

 **Value** As Int

 **Visible** As Boolean

 **Width** As Int

Members description:

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

Color As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

Enabled As Boolean

Height As Int

Initialize (EventName As String)

Invalidate

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

Invalidate2 (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

Invalidate3 (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

IsInitialized As Boolean

Left As Int

Max As Int

Gets or sets the maximum allows value.

RemoveView

Removes this view from its parent.

RequestFocus As Boolean

Tries to set the focus to this view.

Returns True if the focus was set.

SendToBack

Changes the Z order of this view and sends it to the back.

SetBackgroundImage (Bitmap As android.graphics.Bitmap)

SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)

Changes the view position and size.

 Tag As Object

Gets or sets the Tag value. This is a place holder which can used to store additional data.

 Top As Int **Value As Int**

Gets or sets the current value.

 Visible As Boolean **Width As Int**

Gets or sets the view's width.

2.13 Spinner



A folded list that opens when the user clicks on it and allows the user to choose an item. Similar to WinForms ComboBox.

The ItemClick event is raised each time a user presses on an item (even if it is the already selected item).

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

ItemClick (Position As Int, Value As Object)

Members:

 **Add** (Item As String)

 **AddAll** (List As List)

 **Background** As android.graphics.drawable.Drawable

 **BringToFront**

 **Clear**

 **Color** As Int [write only]

 **Enabled** As Boolean

 **GetItem** (Index As Int) As String

 **Height** As Int

 **IndexOf** (value As String) As Int

-  [Initialize](#) (EventName **As String**)
-  [Invalidate](#)
-  [Invalidate2](#) (Rect **As** android.graphics.Rect)
-  [Invalidate3](#) (Left **As Int**, Top **As Int**, Right **As Int**, Bottom **As Int**)
-  [IsInitialized](#) **As Boolean**
-  [Left](#) **As Int**
-  [Prompt](#) **As String**
-  [RemoveAt](#) (Index **As Int**)
-  [RemoveView](#)
-  [RequestFocus](#) **As Boolean**
-  [SelectedIndex](#) **As Int**
-  [SelectedItem](#) **As String** [read only]
-  [SendToBack](#)
-  [SetBackgroundImage](#) (Bitmap **As** android.graphics.Bitmap)
-  [SetLayout](#) (Left **As Int**, Top **As Int**, Width **As Int**, Height **As Int**)
-  [Size](#) **As Int** [read only]
-  [Tag](#) **As Object**
-  [Top](#) **As Int**
-  [Visible](#) **As Boolean**
-  [Width](#) **As Int**

Members description:

Add (Item **As String)**

Adds an item.

Example:

```
Spinner1.Add("Sunday")
```

AddAll (List **As List)**

Adds multiple items.

Example:

```
Spinner1.AddAll(Array As String("Sunday", "Monday", ...))
```

 Background As `android.graphics.drawable.Drawable`

Gets or sets the background drawable.

 BringToFront

Changes the Z order of this view and brings it to the front.

 Clear

Clears all items.

 Color As `Int` [write only]

Sets the background of the view to be a `ColorDrawable` with the given color.

If the current background is of type `GradientDrawable` or `ColorDrawable` the round corners will be kept.

 Enabled As `Boolean` **GetItem (Index As `Int`) As `String`**

Returns the item at the specified index.

 Height As `Int` **IndexOf (value As `String`) As `Int`** **Initialize (EventName As `String`)** **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 Invalidate2 (`Rect` As `android.graphics.Rect`)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 Invalidate3 (Left As `Int`, Top As `Int`, Right As `Int`, Bottom As `Int`)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 IsInitialized As `Boolean` **Left As `Int`** **Prompt As `String`**

Gets or sets the title that will be displayed when the spinner is opened.

 RemoveAt (Index As `Int`)

Removes the item at the specified index.

RemoveView

Removes this view from its parent.

RequestFocus *As Boolean*

Tries to set the focus to this view.
Returns True if the focus was set.

SelectedIndex *As Int*

Gets or sets the index of the selected item. Returns -1 if no item is selected.

SelectedItem *As String* [read only]

Returns the value of the selected item.

SendToBack

Changes the Z order of this view and sends it to the back.

SetBackgroundImage (*Bitmap* *As* *android.graphics.Bitmap*)

SetLayout (*Left* *As Int*, *Top* *As Int*, *Width* *As Int*, *Height* *As Int*)

Changes the view position and size.

Size *As Int* [read only]

Returns the number of items.

Tag *As Object*

Gets or sets the Tag value. This is a place holder which can used to store additional data.

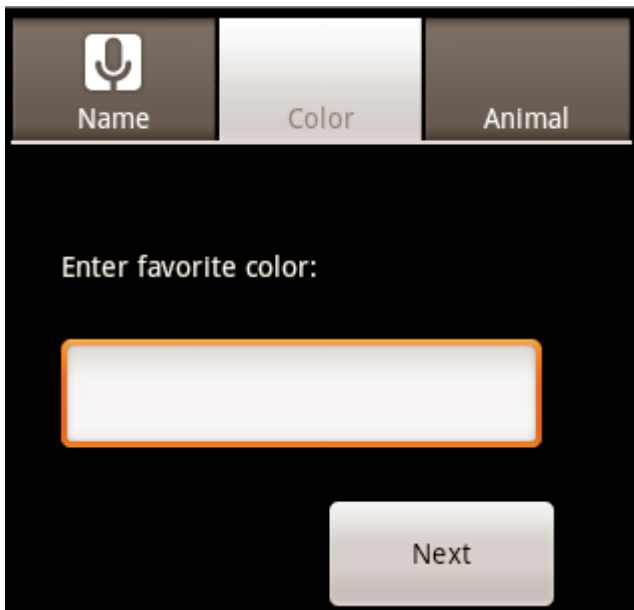
Top *As Int*

Visible *As Boolean*

Width *As Int*

Gets or sets the view's width.

2.14 TabHost



TabHost is a view that contains multiple tab pages. Each tab page contains other child views. See the [TabHost tutorial](#) for more information. This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

TabChanged
Click
LongClick

Members:

[AddTab](#) (Title [As String](#), LayoutFile [As String](#))

[AddTab2](#) (Title [As String](#), [View As](#) android.view.View)

[AddTabWithIcon](#) (Title [As String](#), DefaultBitmap [As](#) android.graphics.Bitmap, SelectedBitmap [As](#) android.graphics.Bitmap, LayoutFile [As String](#))

[AddTabWithIcon2](#) (Title [As String](#), DefaultBitmap [As](#) android.graphics.Bitmap, SelectedBitmap [As](#) android.graphics.Bitmap, [View As](#) android.view.View)

[Background](#) [As](#) android.graphics.drawable.Drawable

[BringToFront](#)

[Color](#) [As Int](#) [write only]

[CurrentTab](#) [As Int](#)

[Enabled](#) [As Boolean](#)

[Height](#) [As Int](#)

-  **Initialize** (EventName **As String**)
-  **Invalidate**
-  **Invalidate2** (Rect **As** android.graphics.Rect)
-  **Invalidate3** (Left **As Int**, Top **As Int**, Right **As Int**, Bottom **As Int**)
-  **IsInitialized** **As Boolean**
-  **Left** **As Int**
-  **RemoveView**
-  **RequestFocus** **As Boolean**
-  **SendToBack**
-  **SetBackgroundImage** (Bitmap **As** android.graphics.Bitmap)
-  **SetLayout** (Left **As Int**, Top **As Int**, Width **As Int**, Height **As Int**)
-  **TabCount** **As Int** [read only]
-  **Tag** **As Object**
-  **Top** **As Int**
-  **Visible** **As Boolean**
-  **Width** **As Int**

Members description:

AddTab (Title **As String**, LayoutFile **As String**)

Adds a tab page.

Title - The page title.

LayoutFile - A layout file describing the page layout.

Example:

```
TabHost1.AddTab( "Page 1", "page1.bal")
```

AddTab2 (Title **As String**, View **As** android.view.View)

Adds a tab page.

Title - The page title.

View - The page content. Usually the view should be a panel containing other views.

AddTabWithIcon (Title **As String**, DefaultBitmap **As** android.graphics.Bitmap, SelectedBitmap **As** android.graphics.Bitmap, LayoutFile **As String**)

Adds a tab page. The tab title includes an icon.

Title - The page title.

DefaultBitmap - The icon that will be drawn when the page is not selected.

SelectedBitmap - The icon that will be drawn when the page is selected.

LayoutFile - A layout file describing the page layout.

Example:

```
Dim bmp1, bmp2 As Bitmap
bmp1 = LoadBitmap(File.DirAssets, "ic.png")
bmp2 = LoadBitmap(File.DirAssets, "ic_selected.png")
TabHost1.AddTabWithIcon("Page 1", bmp1, bmp2, "tabpage1.bal")
```

 **AddTabWithIcon2** (Title As String, DefaultBitmap As android.graphics.Bitmap, SelectedBitmap As android.graphics.Bitmap, View As android.view.View)

Adds a tab page. The tab title includes an icon.

Title - The page title.

DefaultBitmap - The icon that will be drawn when the page is not selected.

SelectedBitmap - The icon that will be drawn when the page is selected.

View - The page content. Usually the view should be a panel containing other views.

 **Background** As android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **CurrentTab** As Int

Gets or sets the current tab.

Example:

```
TabHost1.CurrentTab = (TabHost1.CurrentTab + 1) Mod
TabHost1.TabCount 'switch to the next tab.
```

 **Enabled** As Boolean

 **Height** As Int

 **Initialize** (EventName As String)

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (Rect As android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized** **As Boolean**

 **Left** **As Int**

 **RemoveView**

Removes this view from its parent.

 **RequestFocus** **As Boolean**

Tries to set the focus to this view.

Returns True if the focus was set.

 **SendToBack**

Changes the Z order of this view and sends it to the back.

 **SetBackgroundImage** (**Bitmap** **As** **android.graphics.Bitmap**)

 **SetLayout** (**Left** **As Int**, **Top** **As Int**, **Width** **As Int**, **Height** **As Int**)

Changes the view position and size.

 **TabCount** **As Int** [read only]

Returns the number of tab pages.

 **Tag** **As Object**

Gets or sets the Tag value. This is a place holder which can used to store additional data.

 **Top** **As Int**

 **Visible** **As Boolean**

 **Width** **As Int**

Gets or sets the view's width.

2.15 ToggleButton



A ToggleButton view. This view which is similar to a button has two modes: ON and OFF. When the user presses on it, it will change its mode.

You can set the text with the TextOn and TextOff properties.

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

CheckedChange(Checked As Boolean)

Members:

-  [Background](#) As android.graphics.drawable.Drawable
-  [BringToFront](#)
-  [Checked](#) As Boolean
-  [Color](#) As Int [write only]
-  [Enabled](#) As Boolean
-  [Gravity](#) As Int
-  [Height](#) As Int
-  [Initialize](#) (EventName As String)
-  [Invalidate](#)
-  [Invalidate2](#) (Rect As android.graphics.Rect)
-  [Invalidate3](#) (Left As Int, Top As Int, Right As Int, Bottom As Int)
-  [IsInitialized](#) As Boolean
-  [Left](#) As Int
-  [RemoveView](#)
-  [RequestFocus](#) As Boolean
-  [SendToBack](#)
-  [SetBackgroundImage](#) (Bitmap As android.graphics.Bitmap)

 **SetLayout** (Left **As Int**, Top **As Int**, Width **As Int**, Height **As Int**)

 **Tag** **As Object**

 **TextColor** **As Int**

 **TextOff** **As String**

 **TextOn** **As String**

 **TextSize** **As Float**

 **Top** **As Int**

 **Typeface** **As** android.graphics.Typeface

 **Visible** **As Boolean**

 **Width** **As Int**

Members description:

 **Background** **As** android.graphics.drawable.Drawable

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Checked** **As Boolean**

 **Color** **As Int** [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** **As Boolean**

 **Gravity** **As Int**

 **Height** **As Int**

 **Initialize** (EventName **As String**)

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (**Rect** **As** android.graphics.Rect)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3 (Left As Int, Top As Int, Right As Int, Bottom As Int)**

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized As Boolean**

 **Left As Int**

 **RemoveView**

Removes this view from its parent.

 **RequestFocus As Boolean**

Tries to set the focus to this view.

Returns True if the focus was set.

 **SendToBack**

Changes the Z order of this view and sends it to the back.

 **SetBackgroundImage (Bitmap As android.graphics.Bitmap)**

 **SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)**

Changes the view position and size.

 **Tag As Object**

Gets or sets the Tag value. This is a place holder which can used to store additional data.

 **TextColor As Int**

 **TextOff As String**

Gets or sets the text that will appear in the OFF mode.

 **TextOn As String**

Gets or sets the text that will appear in the ON mode.

 **TextSize As Float**

 **Top As Int**

 **Typeface As android.graphics.Typeface**

 **Visible As Boolean**

 **Width As Int**

Gets or sets the view's width.

2.16 View

View is a special type of object. You cannot create new View objects. However all other view types can be assigned to a view variable.

This allows you to access the shared properties of all views.

For example this code hides all views of an activity:

```
For i = 0 To Activity.NumberOfViews - 1
    Dim v As View
    v = Activity.GetView(i)
    v.Visible = False
```

Next

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Events:

Click

LongClick

Members:

 **Background** As android.graphics.drawable.Drawable

 **BringToFront**

 **Color** As Int [write only]

 **Enabled** As Boolean

 **Height** As Int

 **Invalidate**

 **Invalidate2** (Rect As android.graphics.Rect)

 **Invalidate3** (Left As Int, Top As Int, Right As Int, Bottom As Int)

 **IsInitialized** As Boolean

 **Left** As Int

 **RemoveView**

 **RequestFocus** As Boolean

 **SendToBack**

 **SetBackgroundImage** (Bitmap As android.graphics.Bitmap)

 **SetLayout** (Left As Int, Top As Int, Width As Int, Height As Int)

 **Tag** As Object

 **Top** As Int

 **Visible** As Boolean

 **Width** As Int

Members description:

 **Background** As `android.graphics.drawable.Drawable`

Gets or sets the background drawable.

 **BringToFront**

Changes the Z order of this view and brings it to the front.

 **Color** As Int [write only]

Sets the background of the view to be a ColorDrawable with the given color.

If the current background is of type GradientDrawable or ColorDrawable the round corners will be kept.

 **Enabled** As Boolean

 **Height** As Int

 **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate2** (**Rect** As `android.graphics.Rect`)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **Invalidate3** (**Left** As Int, **Top** As Int, **Right** As Int, **Bottom** As Int)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 **IsInitialized** As Boolean

 **Left** As Int

 **RemoveView**

Removes this view from its parent.

 **RequestFocus** As Boolean

Tries to set the focus to this view.

Returns True if the focus was set.

SendToBack

Changes the Z order of this view and sends it to the back.

SetBackgroundImage (**Bitmap** As **android.graphics.Bitmap**)

SetLayout (**Left** As **Int**, **Top** As **Int**, **Width** As **Int**, **Height** As **Int**)

Changes the view position and size.

Tag As **Object**

Gets or sets the Tag value. This is a place holder which can used to store additional data.

Top As **Int**

Visible As **Boolean**

Width As **Int**

Gets or sets the view's width.

2.17 WebView



The WebView view uses the internal WebKit engine to display Html pages.

The page displayed can be an online page loaded with LoadUrl or a Html string loaded with LoadHtml.

This is an 'Activity Object', it cannot be declared under Sub Process_Globals.

Permissions:

android.permission.INTERNET

Events:

None

Members:

Back

Background As **android.graphics.drawable.Drawable**

-  [BringToFront](#)
-  [Color](#) As Int [write only]
-  [Enabled](#) As Boolean
-  [Forward](#)
-  [Height](#) As Int
-  [Initialize](#) (EventName As String)
-  [Invalidate](#)
-  [Invalidate2](#) (Rect As android.graphics.Rect)
-  [Invalidate3](#) (Left As Int, Top As Int, Right As Int, Bottom As Int)
-  [IsInitialized](#) As Boolean
-  [JavaScriptEnabled](#) As Boolean
-  [Left](#) As Int
-  [LoadHtml](#) (Html As String)
-  [LoadUrl](#) (Url As String)
-  [RemoveView](#)
-  [RequestFocus](#) As Boolean
-  [SendToBack](#)
-  [SetBackgroundImage](#) (Bitmap As android.graphics.Bitmap)
-  [SetLayout](#) (Left As Int, Top As Int, Width As Int, Height As Int)
-  [Tag](#) As Object
-  [Top](#) As Int
-  [Url](#) As String [read only]
-  [Visible](#) As Boolean
-  [Width](#) As Int
-  [ZoomEnabled](#) As Boolean

Members description:

 Back

Goes back to the previous Url.

 Background As `android.graphics.drawable.Drawable`

Gets or sets the background drawable.

 BringToFront

Changes the Z order of this view and brings it to the front.

 Color As `Int` [write only]

Sets the background of the view to be a `ColorDrawable` with the given color.

If the current background is of type `GradientDrawable` or `ColorDrawable` the round corners will be kept.

 Enabled As `Boolean` **Forward**

Goes forward to the next Url.

 Height As `Int` **Initialize (`EventName As String`)** **Invalidate**

Invalidates the whole view forcing the view to redraw itself.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 Invalidate2 (`Rect As android.graphics.Rect`)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 Invalidate3 (`Left As Int`, `Top As Int`, `Right As Int`, `Bottom As Int`)

Invalidates the given rectangle.

Redrawing will only happen when the program can process messages. Usually when it finishes running the current code.

 IsInitialized As `Boolean` **JavaScriptEnabled As `Boolean`**

Gets or sets whether JavaScript is enabled.

JavaScript is enabled by default.

 Left As `Int` **LoadHtml (`Html As String`)**

Loads the given Html.

Example:

```
WebView1.LoadHtml("<html><body>Hello world!</body></html>")
```

You can use "file:///android_asset" to access files added with the file manager:

```
WebView1.LoadHtml("<html><body><img  
src='file:///android_asset/someimage.jpg' /></body></html>")
```

Note that files added with the file manager should be accessed with a lower cased name.

LoadUrl (Url As String)

Loads the given Url.

Example:

```
WebView1.LoadUrl("http://www.google.com")
```

RemoveView

Removes this view from its parent.

RequestFocus As Boolean

Tries to set the focus to this view.

Returns True if the focus was set.

SendToBack

Changes the Z order of this view and sends it to the back.

SetBackgroundImage (Bitmap As android.graphics.Bitmap)

SetLayout (Left As Int, Top As Int, Width As Int, Height As Int)

Changes the view position and size.

Tag As Object

Gets or sets the Tag value. This is a place holder which can used to store additional data.

Top As Int

Url As String [read only]

Returns the current Url.

Visible As Boolean

Width As Int

Gets or sets the view's width.

ZoomEnabled As Boolean

Gets or sets whether the internal zoom feature is enabled.

The zoom feature is enabled by default.

3 Collections

List of types:

[List](#)

[Map](#)

3.1 List

Lists are similar to dynamic arrays. You can add and remove items from a list and it will change its size accordingly.

A list can hold any type of object. However if a list is declared as a process global object it cannot hold activity objects (like views).

Basic4android automatically converts regular arrays to lists. So when a List parameter is expected you can pass an array instead.

For example:

```
Dim List1 As List
List1.Initialize
List1.AddAll(Array As Int(1, 2, 3, 4, 5))
```

Use the Get method to get an item from the list.

Lists can be saved and loaded from files using File.WriteList and File.ReadList.

You can use a For loop to iterate over all the values:

```
For i = 0 To List1.Size - 1
    Dim number As Int
    number = List1.Get(i)
    ...
Next
```

Events:

None

Members:

 [Add](#) (item As Object)

 [AddAll](#) (List As List)

 [AddAllAt](#) (Index As Int, List As List)

 [Clear](#)

 [Get](#) (Index As Int) As Object

 [IndexOf](#) (Item As Object) As Int

 [Initialize](#)

 [Initialize2](#) (Array As List)

 [InsertAt](#) (Index As Int, Item As Object)

💡 **IsInitialized** As Boolean

💡 **RemoveAt** (Index As Int)

💡 **Set** (Index As Int, Item As Object)

📄 **Size** As Int [read only]

💡 **Sort** (Ascending As Boolean)

💡 **SortCaseInsensitive** (Ascending As Boolean)

💡 **SortType** (FieldName As String, Ascending As Boolean)

Members description:

💡 **Add** (item As Object)

Adds an item at the end of the list.

💡 **AddAll** (List As List)

Adds all elements in the specified collection to the end of the list.

Note that you can add an array directly.

Example:

```
List.AddAll(Array As String("value1", "value2"))
```

💡 **AddAllAt** (Index As Int, List As List)

Adds all elements in the specified collection starting at the specified index.

💡 **Clear**

Removes all the items from the list.

💡 **Get** (Index As Int) As Object

Gets the item in the specified index. The item is not removed from the list.

💡 **IndexOf** (Item As Object) As Int

Returns the index of the specified item, or -1 if it was not found.

💡 **Initialize**

Initializes an empty list.

Initialize2 (Array As List)

Initializes a list with the given values. This method should be used to convert arrays to lists. Note that if you pass a list to this method then both objects will share the same list, and if you pass an array the list will be of a fixed size. Meaning that you cannot later add or remove items.

Example:

```
Dim List1 As List
List1.Initialize2(Array As Int(1, 2, 3, 4, 5))
```

Example:

```
Dim List1 As List
Dim SomeArray(10) As String
'Fill array...
List1.Initialize2(SomeArray)
```

InsertAt (Index As Int, Item As Object)

Inserts the specified Item in the specified index. As a result all items with index larger then the specified index are shifted.

IsInitialized As Boolean

RemoveAt (Index As Int)

Removes the item at the specified index.

Set (Index As Int, Item As Object)

Replaces the current item in the specified index with the new item.

Size As Int [read only]

Returns the number of items in the list.

Sort (Ascending As Boolean)

Sorts the list.

The items must all be numbers or strings.

SortCaseInsensitive (Ascending As Boolean)

Lexicographically sorts the list, ignoring the characters case.

The items must all be numbers or strings.

SortType (FieldName As String, Ascending As Boolean)

Sorts a list with items of user defined type. The list is sorted based on the specified field.

FieldName - The case-sensitive field name that will be used for sorting. Field must contain numbers or strings.

Ascending - Whether to sort ascending or descending.

Example:

```
Sub Process_Globals
    Type Person(Name As String, Age As Int)
End Sub

Sub Activity_Create(FirstTime As Boolean)
    Dim Persons As List
    Persons.Initialize
    For i = 1 To 50
        Dim p As Person
        p.Name = "Person" & i
        p.Age = Rnd(0, 121)
        Persons.Add(p)
    Next
    Persons.SortType("Age", True) 'Sort the list based on the Age field.
    For i = 0 To Persons.Size - 1
        Dim p As Person
        p = Persons.Get(i)
        Log(p)
    Next
End Sub
```

3.2 Map

A collection that holds pairs of keys and values. The keys are unique. Which means that if you add a key/value pair (entry) and the collection already holds an entry with the same key, the previous entry will be removed from the map.

Fetching an item is done by looking for its key. This is usually a very fast operation ($O(1)$ compared to $O(n)$ in a list).

The key should be a string or a number. The value can be any type of object.

Note that this map implementation does return items in the same order as they were added.

Usually you will use Put to add items and Get or GetDefault to get the values based on the key.

If you need to iterate over all the items you can use a For loop like:

```
For i = 0 To Map1.Size - 1
    Dim Key, Value As String
    Key = Map1.GetKeyAt(i)
    Value = Map1.GetValueAt(i)
Next
```

Similar to a list, a map can hold any object, however if it is a process global variable then it cannot hold activity objects (like views).

Maps are very useful for storing applications settings.

You can save and load maps with File.WriteMap and File.ReadMap.

Events:

None

Members:

 [Clear](#)

 [ContainsKey](#) (Key As Object) As Boolean

 [Get](#) (Key As Object) As Object

 [GetDefault](#) (Key As Object, Default As Object) As Object

 [GetKeyAt](#) (Index As Int) As Object

 [GetValueAt](#) (Index As Int) As Object

 [Initialize](#)

 [IsInitialized](#) As Boolean

 [Put](#) (Key As Object, Value As Object) As Object

 [Remove](#) (Key As Object) As Object

 [Size](#) As Int [read only]

Members description: **Clear**

Clears all items from the map.

 ContainsKey (Key As Object) As Boolean

Tests whether there is an item with the given key.

Example:

```
If Map.ContainsKey("some key") Then ...
```

 Get (Key As Object) As Object

Returns the value of the item with the given key.

 GetDefault (Key As Object, Default As Object) As Object

Returns the value of the item with the given key. If no such item exists the specified default value is returned.

 GetKeyAt (Index As Int) As Object

Returns the key of the item at the given index.

GetKeyAt and GetValueAt should be used to iterate over all the items.

These methods are optimized for iterating over the items in ascending order.

Example:

```
For i = 0 to Map.Size - 1
    Log("Key: " & Map.GetKeyAt(i))
    Log("Value: " & Map.GetValueAt(i))
Next
```

 GetValueAt (Index As Int) As Object

Returns the value of the item at the given index.

GetKeyAt and GetValueAt should be used to iterate over all the items.

These methods are optimized for iterating over the items in ascending order.

Example:

```
For i = 0 to Map.Size - 1
    Log("Key: " & Map.GetKeyAt(i))
    Log("Value: " & Map.GetValueAt(i))
Next
```

 Initialize

Initializes the object.

Example:

```
Dim Map1 As Map
Map1.Initialize
```

 IsInitialized As Boolean

Put (Key As Object, Value As Object) As Object

Puts a key/value pair in the map, overwriting the previous item with this key (if such exists).

Returns the previous item with this key or null if there was no such item.

Note that if you are using strings as the keys then the keys are case sensitive.

Example:

```
Map1.Put("Key", "Value")
```

Remove (Key As Object) As Object

Removes the item with the given key, if such exists.

Returns the item removed or null if no matching item was found.

Size As Int [read only]

Returns the number of items stored in the map.